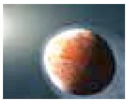
Protective parent frog
Nest building behaviours were noticed in the Goliath frog who heave rocks to create a sort-of nest. **https://dgit.in/sep19-15**



Football-shaped planet is leaking...
Metals! The Hubble spotted this planet, which is sometimes 10 times hotter than any other world explored yet. **https://dgit.in/sep19-16**

and a researcher at KAIST and the University of Colorado Boulder. He believes that developing gadgets which marry neural stimulation with diagnostic utility can enable the creation of 'cyborg-like' brain boosting devices someday, which can maintain healthy neural function and possibly even enhance neural aptitude.

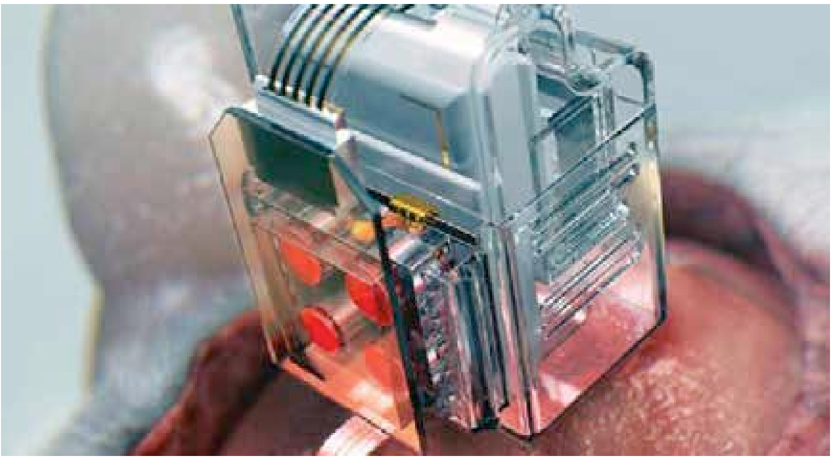
HOW IT WORKS

The device houses replaceable drug cartridges resembling Legos and Bluetooth low-energy. The cartridges deploy the specific drug to neurons of interest while Bluetooth low-energy enables the usage of drug and light therapy over long periods of time since it is power efficient. Raza Qazi stated that this particular method overshadows one used by neuroscientists before since they usually involve solid metal tubes and optical fibres to deliver drugs and light. The rigidity not only limits the subject's movement due to being tethered to bulky equipment but also causes a lesion in the soft brain tissue over a prolonged time. This approach has seen some minor improvements to mitigate adverse tissue response by adding soft probes and wireless platforms, however, they are still limited by their inability to deliver drugs over extended periods of time and their bulky and complex architecture.

On the other hand, the device created by KAIST researchers achieves wireless drug delivery efficiently since it solves the challenge of exhaustion and evaporation of drugs. The replaceable drug cartridge system allows neuroscientists to study the brain circuits for several months on end without the patients having to go through the harrowing experience of drugs running out frequently.

You could call these Lego-like cartridges a sort of 'plug-n-play' cartridge that is deposited into a brain implant with a soft, ultrathin probe (which has the thickness of a single human hair). This probe consists of microfluid channels and tiny LEDs (that are smaller than a grain of salt), which allow unlimited drug doses and light delivery. This treatment has only been tested on mice so far, however, human tests are starting very soon.

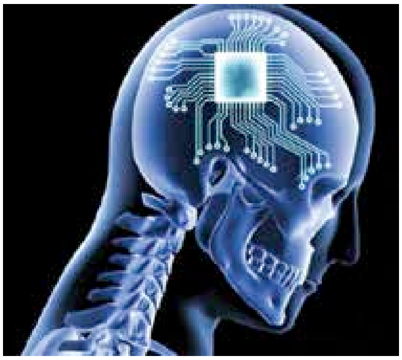
The probe is also capable of administrating four different drugs and two different light wavelengths inside the brain tissues of mice. The neural device has also been designed to be optofluidic, meaning its pumps are thermally actuated for efficient drug delivery. After the brain implant has been fitted securely deep inside the brain tissue, researchers can wirelessly program as well as control the neural device's operation, the condition of the heaters, and the LED light wavelength of blue or orange by simply using the associated smartphone app which the researchers have developed. The wireless range is extremely long as well - anywhere between 10 to 100 metres - with omnidirectional access.



This wireless device can manipulate brain cells with a smartphone

USE CASES

Qazi stated, "The wireless neural device enables chronic chemical and optical neuromodulation that has never been achieved before". A professor of anesthesiology, pain medicine and pharmacology at the University of Washington clarified, "It (the device) allows us to better dissect the neural circuit basis of behaviour, and how specific neuromodulators in the brain tune behaviour in various ways. We are also eager to use the device for



A brain implant controlled by a smartphone manipulates brain cells

complex pharmacological studies, which could help us develop new therapeutics for pain, addiction, and emotional disorders."

The researchers believe that this device, and others that stem from the concepts and innovations brought to life in the device, can immensely benefit in demystifying the brain and its plethora of diseases. In the future, neuroscience laboratories and pharmaceutical companies can use this device to leverage their research throughput by cutting down the time it takes to determine drug effectiveness and possible side effects by using the same test subjects over prolonged periods. It also trimmed down the number of animal subjects that would be required to conduct these tests.

While this innovation may not satiate your desires to control your smartphone using your mind, it does possess the power to spark innovation and treat brain-related ailments in a more cost and time productive manner in the future.